

Arnav Mangaonkar

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EDUCATION

SIES Graduate School of Technology

Bachelor of Engineering in Artificial Intelligence and Machine Learning

Navi-Mumbai, MH

2022-2026

EXPERIENCE

Data Analyst Intern

Dec 2024 – Mar 2025

Tata Power Renewables Limited

- Developed a web application that can analyze the energy data and plot it dynamically with reference to time and other parameters
- Developed a program that can analyze and monitor real-time energy generation data and flag malfunctioning or anomaly in transformers

Embedded Developer Consultant

Jun 2025 – Jan 2026

Vektor3d Systems LLP

- Developing a detection and alert system for anomalies and errors occurring in additive manufacturing specifically 3d printing

PROJECTS

ByteCave - Self Hosted DNS, NAS, VPN, Ad-Blocker, Media Server

Nov 2025 - Present

- Deployed Pi-hole DNS filtering with WireGuard VPN on Raspberry Pi for secure ad-blocking and remote access.
- Set up a self-hosted NAS with role-based access control and secure file sharing across local and VPN clients.
- Optimized DNS resolution/caching and enforced firewall rules to strengthen privacy, security, and performance.
- Deployed PiCone a lightweight custom-made media server for 32-bit ARM based CPUs.

GateNet-AM | *Gated NN for anomaly detection in additive manufacturing*

Jun 2025 – Feb 2026

- Built a real-time anomaly detection pipeline for FDM 3D printing using video input to identify print failures early.
- Designed **GateNet-AM**, a gated neural network architecture optimized for lightweight, low-latency edge inference.
- Implemented automated defect classification with visual annotations and logging for performance tracking and debugging.

CodeComp | *Repository-Aware Codebase Q&A Assistant*

Jan 2026 – Apr 2026

- Built a Flask app that clones GitHub repositories, parses project structure, and enables interactive codebase exploration.
- Implemented hybrid understanding with Python AST chunking and text fallback, then generated semantic embeddings using BAAI/bge-small-en-v1.5.
- Developed hybrid retrieval (cosine + keyword scoring) and integrated Groq-based JSON-grounded answers with chat history and debug tracing.

PUBLICATIONS / IP

MotoGuard (Copyright) | *Car Anti-Theft System*

Published 2025

- Implemented a driver-side camera module for intruder identification and evidence capture.
- Developed an MJPEG encoder on ESP32 to optimize bandwidth and enable low-latency video streaming.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, Embedded C, SQL (PostgreSQL, MySQL), HTML/CSS

Frameworks: Flask, Streamlit, Tkinter, Swing

Libraries: Pandas, NumPy, Matplotlib, BeautifulSoup, Scikit-learn, OpenCV, ultralytics

Skills: Computer Networks, Embedded Linux, Raspberry Pi, AWS, Version Control, Embedded Firmware, Concurrency, Machine Learning, Computer Vision, Object Detection, Edge Deployment, IoT, RAG Pipelines, Vector Databases, Generative AI, LLM Applications

Spoken Languages: English, Marathi, Hindi